

Wearable Driver Monitoring System

Michael Benusa

Department of Electrical and Computer Engineering
University of St. Thomas
St. Paul, MN, USA
mabenusa@stthomas.edu

Cheol-Hong Min

Department of Electrical and Computer Engineering
University of St. Thomas
St. Paul, MN, USA
cmin@stthomas.edu

Abstract— *The proposed system measures the attentiveness of the driver using the developed wearable device with electrodes placed on the temples of the driver. By using EEG signals, the accuracy of attentiveness detection may be increased compared to standalone camera-based systems. Preprocessing was used to remove noise from the measured signals. Features were extracted from the signals, and an optimal feature set was determined by analyzing the feature's effectiveness. A Support Vector based machine-learning algorithm was used to classify the driver's states. To gather data for each state, experiments were conducted on an indoor driving simulator with four subjects. The proposed model's classification accuracy was 84.4% for the five-fold cross-validation of the data. In a subject-independent test, the accuracy was 84.2% for the four states (city attentive, city distracted, highway attentive, and highway distracted). When adding a third classifier category to detect drowsiness from inattentiveness, the proposed model's accuracy was dropped to 72.6%. In this preliminary study, abnormal conditions of the drivers were detected and classified using the proposed system.*

Keywords—driver condition monitoring, EEG, biosensing, detection, classification

I. INTRODUCTION

Driver safety is an issue of great importance. The increasing level of technology is bringing more distractions to drivers on the road. More than 3,000 people were killed in the United States, and over 400,000 were injured due to distracted driving. 1 in 5 people who died in those accidents was not in vehicles. They were walking, riding bikes or outside a vehicle [1]. Distracted driving greatly affects people both on and off the road. Therefore, a system that can determine and monitor the level of attentiveness of a driver could be very useful. Various methods have been used to monitor the drivers of vehicles in recent years. Some researchers have developed systems condensed to fit the shape of a wristwatch [2], monitoring accelerometer data corresponding to movement and measurements of skin conductance and sweat production to monitor drivers' stress levels. This system provides a vibration to the user to alert them of inactive driving behavior [3]. Other researchers have monitored drivers with cameras around the vehicle to track eye and head movement [4-6]. A lack of eye or

head movement would allow the researcher to recognize periods of drowsiness in the driver of the vehicle. Conversely, an increased eye and head movement from the norm could indicate a distracted driver. An issue with this approach is that camera-based methods require special installation within a vehicle, and these systems need to perform better in low-light settings or with reflective/shaded eyewear. For this reason, a wearable approach is preferred.

Our initial research aimed to diagnose driver attentiveness through an electroencephalogram (EEG). This allows various frequency band powers to be extracted and monitored to observe changes in attentive states. An algorithm was developed using MATLAB [16] to establish parameters defining the band powers to determine if distracted driving has occurred.

EEG signals were measured using a Ganglion biosensing board with Open BCI's software. Various tests were conducted to observe the changes in EEG characteristics depending on whether a driver was attentive. For safety reasons, these tests were conducted using an online open-world driving simulator containing normal traffic signals (stop lights, speed limits, traffic signs), various road types (city, highway, country roads), and regular computer-simulated traffic. These collected signals were then processed using MATLAB.

II. DATA COLLECTION

A. Devices

The Open BCI Ganglion biosensing board was used to measure the EEG signals. This is a compact device with four channels intended to measure differential biosensing signals for EEG, electromyography (EMG), electrocardiography (ECG), and accelerometer data (ACC). This board is powered by four AA alkaline batteries, making the system easily powered and portable. Four gold cup electrodes were placed on the participants using Ten20 conductive paste, which is safe for participants and can be easily removed, which can measure the EEG signals with the Ganglion board. These gold cup electrodes were placed on the positive terminals of the input channel pins of the board. An image of the board used along with the power supply is shown in Fig. 1. The Ganglion board was used in conjunction with OpenBCI_GUI software for PC applications. This easy-to-use real-time collection software includes multiple

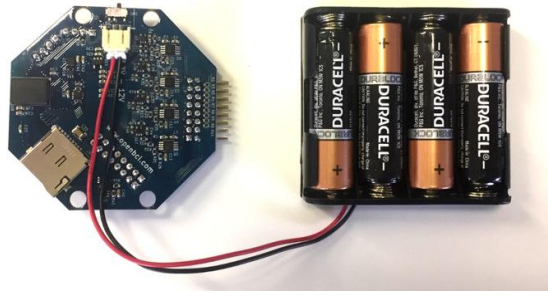


Fig. 1. Ganglion Biosensing board and used power supply for EEG data collection.

real-time options for viewing signal characteristics. These EEG measurements were on the order of hundreds of microvolts. This software also allows the user to set the sampling frequency of the system. Since most frequency components of interest in an EEG were from theta waves [4-8 Hz] to beta waves [13-35 Hz], the desired appropriate sampling rate was 200 Hz. The system also allows for real-time digital filtering of the signals. However, these methods were not used since any filtering could be done after collecting the entirety of the signal in MATLAB. The software stores each trial's data in a text file. These files were manually imported into the MATLAB platform to manipulate and view the data.

B. Electrode Placement

Electrode placement for EEG signal was chosen to collect data with the least amount of noise and in a location that would be easy to access in a usable real-world system. Previous research was completed by collecting data on the occipital lobe. Still, this sensor location was deemed improbable for a real-world scenario because the hair on the back of the head makes it hard for the sensors to stay attached soundly to the skin. The frontal lobe was chosen to measure the participants' EEG to identify attentiveness in a way that would be possible for daily use. This allowed for easy placement of gold cups only in locations with direct contact with the participant's skin for optimal signal collection. Four cups in total were used to collect EEG signals. A reference cup was placed on the participant's ear lobe, and two positive terminal channels were placed just above the participant's eyebrows, centered directly above the center of the participant's eyes. The head movement and wire movement introduced much noise into the system, so a fourth cup was placed on the opposite earlobe and on the D_G channel's positive terminal. This pin is used to reduce signal noise, as any head movements that cause similar noise on all channels were measured on the D_G channel and removed from the final output. This left eye movement was the only signal noise the sensors picked up. It was found that using a wider cup placement (far temples or above far ends of eyebrows) picked up more electrical signals from eye movements and EMG blinking artifacts than centered above the eyebrows did. Ear lobes were chosen as low noise reference points for the ref pin and the D_G pin because there were no muscles to cause EMG artifacts, and relatively, no ECG artifacts were included.

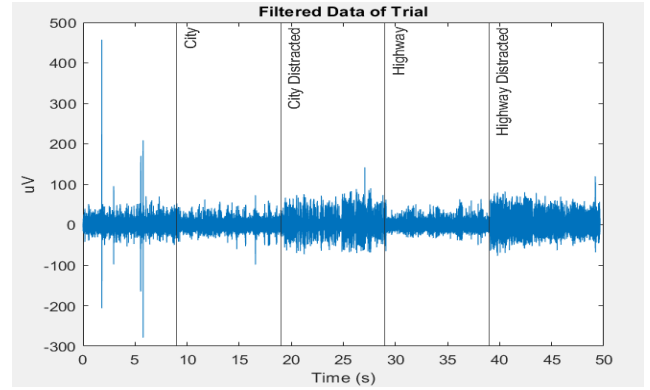


Fig. 2. Example of filtered EEG recording. Each run consists of 10 min of setup, city, city distracted, highway, and highway distracted driving.

III. FEATURE EXTRACTION

Once the collected signals had been imported into MATLAB, they were filtered to reduce noise in the system. An ideal bandpass filter was applied using the bandpass function to only allow signal frequencies from 4 Hz to 50 Hz, shown in Fig. 2. This also removes any power system's 60 Hz noise, baseline wander, or DC offsets from the signal. The most undesired noise remains from blinking artifacts or eye movements already reduced using the real-time systems D_G pin.

The first part of the data was from 0 to 10 minutes; the system was turned on during this setup period, and the driving simulator was set up. The first data related to setup was removed from the final algorithm design. In the second part of the dataset, from 10 to 20 minutes, the participant was told to drive on city roads in the simulator. They were told to practice good driving by following any traffic signals and the flow of traffic created by the computer-generated traffic. In the third part of the dataset, from 20 to 30 minutes, the driver was told to remain in the same city area but now was told to take out their phone and either engage in texting conversations or use social media apps (it was found that either activity produces similar results) while still trying to follow traffic rules. This led to some in-game accidents, which the participant was told to ignore and return to driving so that the system could continue collecting data. The final algorithm was designed to detect distracted driving rather than crash events, so any events in the simulator were unwanted data. From 30 to 40 minutes, the participant was told to find the nearest highway to their current location in the simulator and drive attentively, following traffic rules and flow. From 40 to 50 minutes, the participant was told to take out their phone again and engage in social media or texting conversations while trying to pay attention to driving on the highway.

These four categories of interest were City Driving (10-20 minutes), City Distracted Driving (20-30 minutes), Highway Driving (30-40 minutes), and Highway Distracted Driving (40-50 minutes). It was hypothesized that both distracted scenarios would have the majority of band power in the lower frequency

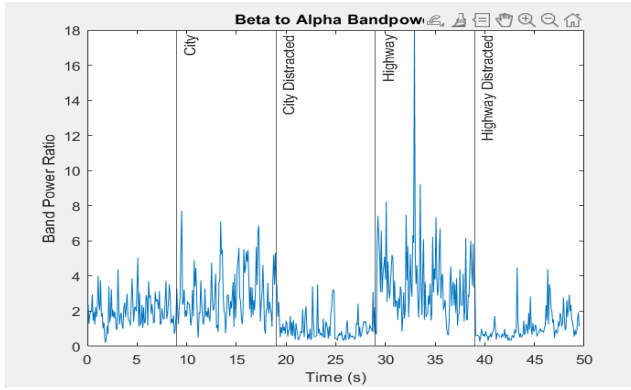


Fig. 3. Beta to Alpha band power ratio for different driving conditions.

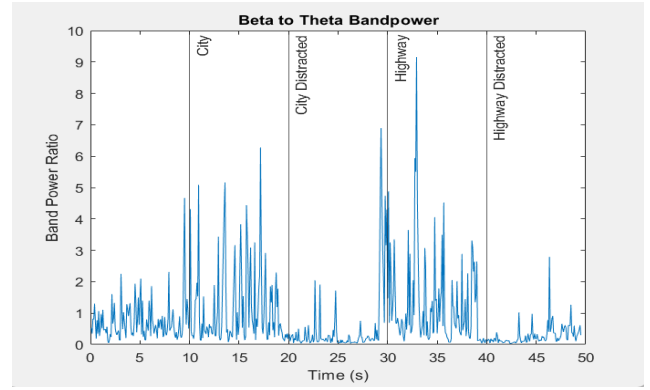


Fig. 4. Beta to Theta band power ratio for different driving conditions.

spectrum compared to attentive driving. It was also hypothesized that both highway categories might have more frequency band powers in lower frequency ranges than the city driving scenarios since fewer driving events, such as stopping, starting, turning, or merging events, were required. Features were extracted using one-second windows with 50% overlap. This allows for a new classification of events every half a second and a sufficient number of samples to be included in every feature calculation to extract a meaningful estimation of what the driver was doing. Features extracted were Theta band power (8-7 Hz), Alpha band power (8-13 Hz) [7], and Beta band power (13-35 Hz). Two ratios were also used: Beta-to-Alpha ratio and Beta-to-Theta ratio (see Fig. 3, 4). These ratios have also been the areas of study for frontal lobe state monitoring in other research projects [8]. The frequency band power collection shows that most signal power generally occurs in the lower frequency spectrum. The Theta band power has large spikes during distracted driving scenarios for highway and city driving. There were still spikes in the Theta band for city and highway attentive driving scenarios, but these spikes were much less frequent and lower in magnitude. These spikes could be caused by the participant looking at their cellular device, entering, and ending a distracted period with the participant looking back to the road to gauge their current position on the road and traffic conditions.

Beta-to-Alpha ratios show a slight increase in the ratio value for city and highway driving attentive scenarios. The band power ratios were lower for distracted driving. This data collection shows that during distracted driving events, there was more signal power in the lower-frequency spectrums and less in the high-frequency spectrums. Beta-to-Theta ratios also show an increased ratio value for city and highway driving attentive scenarios. This increase was much more prominent than the Beta-to-Alpha ratio but less consistent for distract scenarios. The band power ratios were lower for distracted driving. This data collection shows that during distracted driving events, there was more signal power in the lower-frequency spectrum and less in the high-frequency spectrum. From this data, it was assumed that both band power ratios and even the Theta and Beta raw band powers could be useful in classifying attentive versus distracted driving scenarios.

IV. CLASSIFICATION ALGORITHM

While training and testing the dataset, only attentive and distracted were used as possible states. This was because the most important aspect of the system was that the driver was either attentive and assumed to be driving well, or they were distracted and should be notified to pay more attention. MATLAB was used to develop an algorithm that can be used to classify driving states automatically. All features were used to develop a baseline default setting when testing different algorithms. This default set was tested for data trees, KNN, SVM, neural networks, and linear discriminants. It was found that Quadratic SVM had the highest accuracy of all other models, with an accuracy of 81.9%. This model was further tested using all possible feature selections using the five extracted features described above. It was found that all features were useful in classifying different driver states, and removing any one of them caused a significant decrease in model performance. The final selected model was the Quadratic SVM model with Theta band power, Alpha band power, Beta band power, Beta-to-Theta ratio, and Beta-to-Alpha ratio included. This system had an accuracy of 84.4 %, using a 5-fold cross-validation method within MATLAB's classification learner. This model was then exported into the workspace to allow for testing using separate data sets.

The data was tested using the exported model in MATLAB. When tested against an entirely separate dataset from the training data, the model had an accuracy of 84.2%, nearly identical to the performance of the validation model set. Fig. 5 shows that the classifier can detect the region of distracted driving. If the system were to classify attentive driving as distracted, the consequences could be less severe than if the system classified distracted driving as attentive. Therefore, the overall goal in developing this system should be to try and reduce the false positive rate for attentive driving. A third state was added to further test the algorithm. The system inputs a drowsy sample and the previous distracted and attentive categories. This data was collected late in the evening when the participant was tired. The participant was instructed to sit still with eyes shut as if they had dosed off while driving. It was assumed that this category would have the least Beta wave and most Theta wave, so the system could categorize this as a separate dataset.

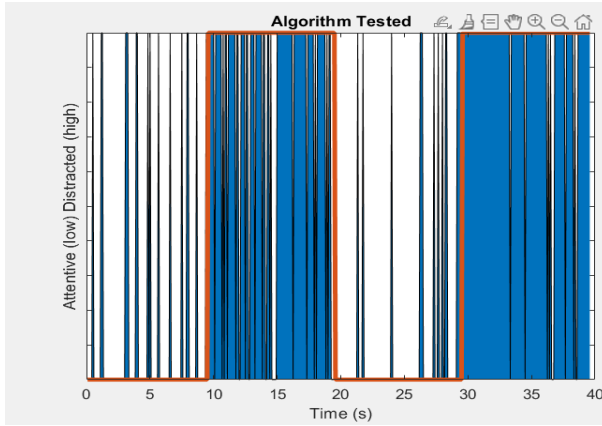


Fig. 5. Algorithm tested using a separate test set. The blue area is distracted classification by the algorithm, and the red line is the ground truth of the test scenario.

When developing the algorithm for this three-state classifier, a similar procedure was followed, where all features were used along with all available machine learning classifiers using MATLAB's Classification Learner. It was found that, again, the Quadratic SVM classifier was still the best-performing model using a five-fold validation of the data input. All features were again found valuable for this classifier. The classifier shows 69.9% accuracy, lower than the previous two-class classifier. It was observed in the confusion matrix that the attentive and drowsy states were similar in performance as before. Still, the drowsy states were mistaken about half the time for the attentive state (Fig. 6). This is of much larger consequence than if the system had mistaken the drowsy state as distracted since the driver would not be alerted for a dangerous driving state.

When validating this model, the model performed better than expected with another testing set. The accuracy of the exported model was 72.6%. Still, over half of the drowsy states need to be classified better than the other two and were the poorest performing of any conditions. It is believed that drowsy must be too similar to an attentive driving state with little noise introduced from head movements and eye movements. It is also possible that participants were wandering while collecting drowsy data since it may be hard to measure the feeling of tiredness as the subject is only imitating tiredness.

V. CONCLUSIONS

This complete system shows that it is possible to detect distracted driving using only EEG data. The best model using only EEG data was a quadratic SVM classifier. The tested accuracy of this classifier was 84.2% EEG data when applied to a complete system, which could include eye tracking, ECG heart rate monitoring and blink rate through EMG signals could further increase the overall system. Prior research also shows the effectiveness of these signals. It is important to note that a while system used for a safety scenario like this should be expected to perform at a much higher overall accuracy than 84.4% when the model was created using only EEG features. A holistic approach should be used to increase accuracy using eye tracking systems, heart rate monitors, and blink rates.

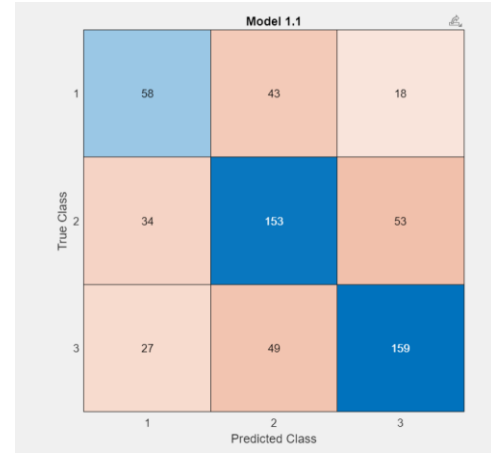


Fig. 6. Quadratic SVM Confusion Matrix for the drowsy state included.

When used for a drowsy, attentive, and distracted test, the model was much less accurate than distracted driving monitoring that used only EEG signals. The classifier was only 72.6% accurate compared to 84.4%. However, other methods for monitoring drowsy driver behavior may be more effective. When a driver becomes drowsy and begins to dose off, the blink and heart rates of the driver alter significantly [2]. These sensors may be better options than an EEG to increase the accuracy of an eye-tracking classifier.

Further research is being conducted to develop a real-time system that detects distracted driving in real-time. The limitation at the time of this study is that the data was collected only using a driving simulator. Real driving scenarios will also include more noise to the system in the form of vibration from the road and movement of the driver while checking mirrors or merging. The data set collected also included four people. At this time, more data is being collected to increase the system's accuracy for people with different habits than those tested.

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Appendix